RAG

AI doučovateľ

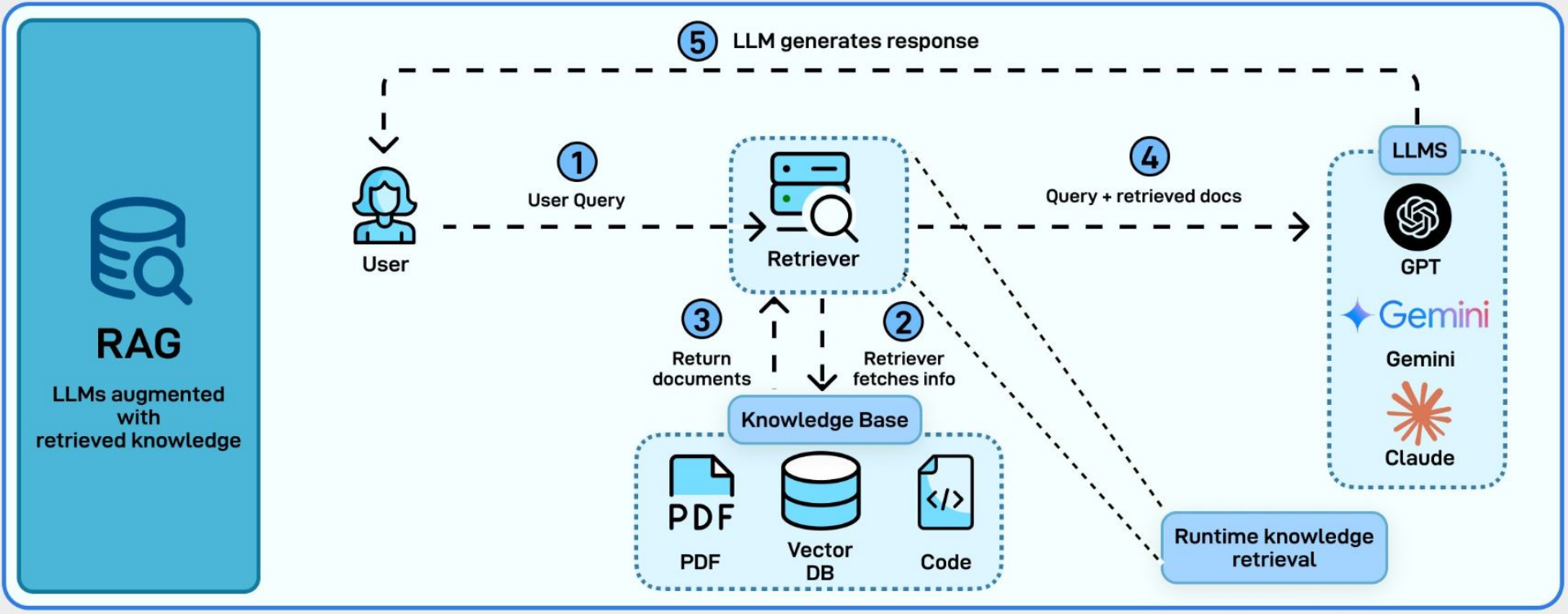
🤖🎓

Kade Mackintosh

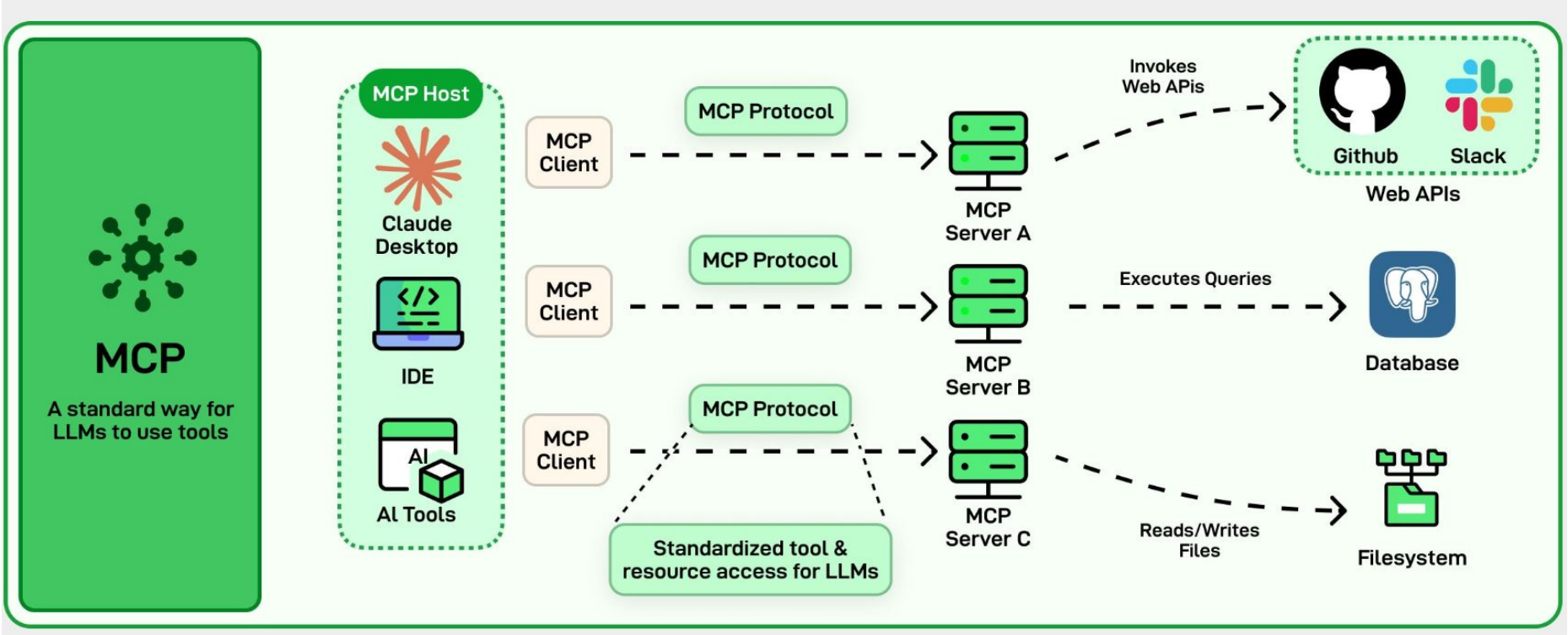
WebUp #65 - 25.3.2026

RAG?

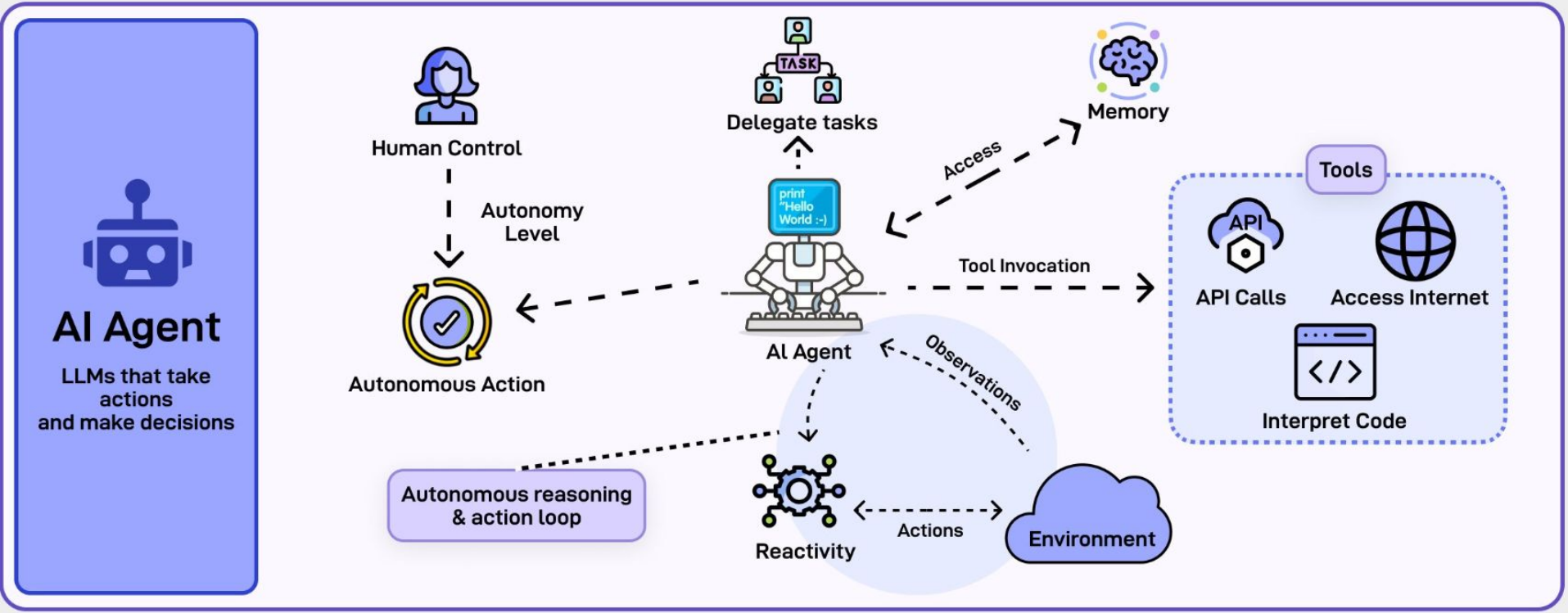
Retrieval **A**ugmented **G**eneration



MCP vs. RAG


MCP
A standard way for LLMs to use tools
MCP Host
Claude Desktop
IDE
AI
AI Tools
MCP Client
MCP Protocol
MCP Client
MCP Protocol
MCP Client
MCP Protocol
Standardized tool & resource access for LLMs
MCP Server A
MCP Server B
MCP Server C
Invokes Web APIs
Executes Queries
Reads/Writes Files
Github
Slack
Web APIs
Database
Filesystem

Agent?


AI Agent
LLMs that take actions and make decisions
Human Control
Autonomy Level
Autonomous Action
Delegate tasks
TASK
print "Hello World :-)
AI Agent
Access
Memory
Tool Invocation
Tools
API
API Calls
Access Internet
Interpret Code
Observations
Autonomous reasoning & action loop
Reactivity
Actions
Environment

Výsledok:

maturitny-tester.sk

maturitny-tester.sk/test/matematika/126306

Spustiť

Časovač

MINISTERSTVO ŠKOLSTVA, VEDY, VÝSKUMU A ŠPORTU SLOVENSKEJ REPUBLIKY

KÓD TESTU
6306

MATURITA 2012
EXTERNÁ ČASŤ

PREČÍTAJTE SI NAJPRV POKYNY K TESTU!
Želáme vám veľa úspechov!

Časť I

Vyriešte úlohy **01** až **20** a do odpoveďového hárka zapíšte vždy **iba výsledok** — nemusíte ho zdôvodňovať ani uvádzať postup, ako ste k nemu dospeli.

Obrázky slúžia len na ilustráciu, nahrádzajú vaše náčrty, dížky a uhly v nich nemusia presne zodpovedať údajom zo zadania úlohy.

1 Rovnica $\sqrt{44-x} = 2 - x$ má práve jeden koreň v množine reálnych čísel. Určte

So súhlasom NIVaM © 2026 maturitný tester | VOP | GDPR

Vysvetliť odpoveď

60.) Before the Harry Potter project, Emma Watson had never ✕ → acted professionally.

Skryť vysvetlenie

Vysvetlenie:

Správna odpoveď je acted. V texte sa uvádza, že Watson mala predtým "never acted professionally", čo naznačuje, že jej prvé herecké vystúpenie bolo v tejto súvislosti významné a úspešné.

Bolo toto vysvetlenie užitočné?

Áno | Nie | Návrh

Skryť chat

Opýtaj sa na čokoľvek

Super, kde presne v texte je to napísané. Uveď celú vetu

25.01.2026 00:12:00

V texte je to napísané v tejto vete: "Watson had never acted professionally when her theatre teachers suggested her to agents looking to cast an upcoming movie based on the first novel of the best-selling Harry Potter series."

25.01.2026 00:12:02

2021 - 2026 © maturitný tester | Kontakt: kontakt@maturitny-tester.sk

VOP | Ochrana osobných údajov | Cookies

Ako na to?

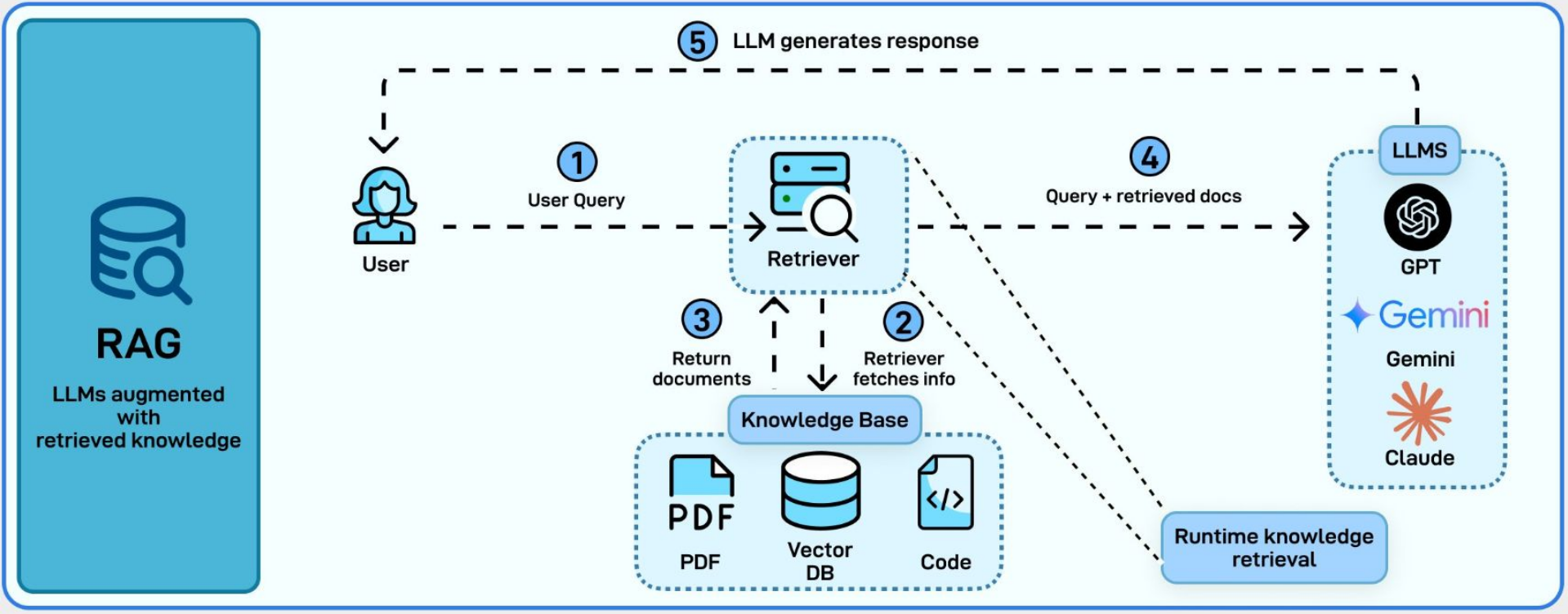
🛠️ Tech:

- Angular
- NestJS
- PostgreSQL + pgvector

- LangChain + OpenAI

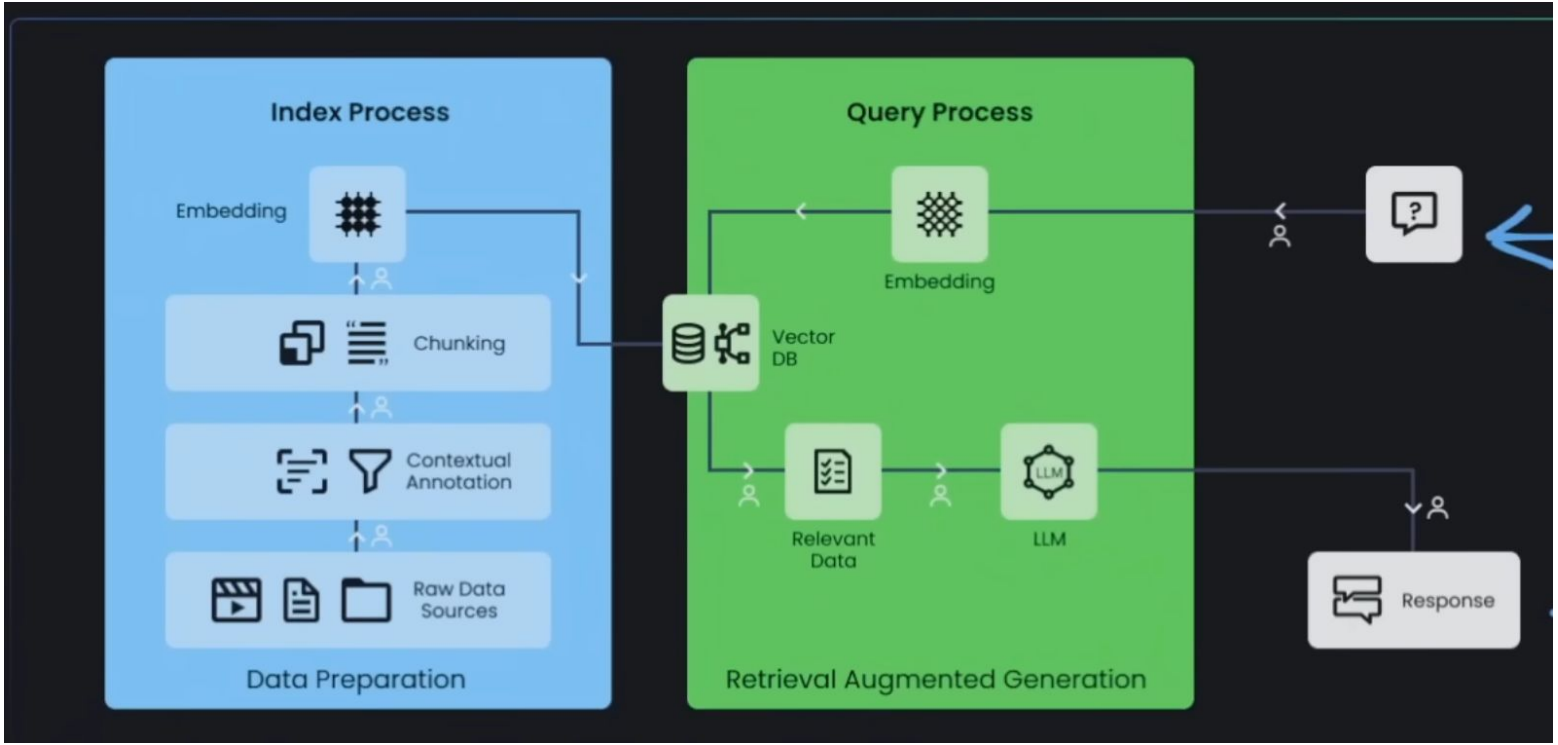
RAG?

Retrieval **A**ugmented **G**eneration



RAG?

Retrieval **A**ugmented **G**eneration



1. Pripraviť DB

Pripraviť PostgreSQL na ukladanie vektorov.

Nainštalovať pgvector extension a vytvoriť tabuľky.

```
CREATE EXTENSION IF NOT EXISTS vector;
```

Explore / pgvector / pgvector

pgvector/pgvector

By pgvector • Updated 26 days ago

Open-source vector similarity search for Postgres

IMAGE

DATABASES & STORAGE

97 50M+

test_embeddings
123 id
A-Z test_name
A-Z question
123 chunk_index
A-Z chunk
embedding

2. Spracovať dáta

- Nahrávky:

- Obrázky:

Part 1: An Interview with Kate Platt, an American Student in Japan (7 points)

Vypočujete si prvú nahrávku. Na základe vypočutého vyberte správnu odpoveď. Vždy je správna iba **jedna** z ponúkaných možností.

Svoje odpovede vyznačte na odpoveďovom hárku označenom piktogramom "X".

Teraz máte dve minúty na prečítanie úloh **01 – 07.**

Audio - ukážka

0:00 / 3:07 1x

2 Vo finále súťaže točí súťažiaci postupne dvoma kolesami šťastia znázornenými na obrázku. Kolesá sú rozdelené na štyri časti: plná, šrafovaná, bodkovaná a prázdna. Percentá v jednotlivých častiach znamenajú pravdepodobnosť vytočenia tejto časti. Súťažiaci vyhrá hlavnú cenu, ak na oboch kolesách vytočí šrafovanú časť. Určte ako číslo z intervalu $\langle 0; 1 \rangle$ pravdepodobnosť, že súťažiaci vyhrá hlavnú cenu.

zadaj odpoveď

🎶Nahrávky

```
pip install openai-whisper

sudo apt install ffmpeg
```

Script (`generate-media-transcripts.ts` **) volá Whisper CLI:**

```
// projects/backend/scripts/generate-media-transcripts.ts

const model = 'medium'; // Options: tiny, base, small, medium, large
const languageFlag = language ? `--language ${language}` : '';

const command = `whisper "${audioPath}" --model ${model} ${languageFlag} --output_format txt --output_dir "${audioDir}"`;

execSync(command, {
  stdio: 'pipe',
  encoding: 'utf-8',
});
```

- **Model:** `medium` (1.5 GB) — dobrá presnosť pre vzdelávací obsah
- **Language:** Vždy nastaviť manuálne! (`en` pre angličtinu, `sk` pre slovenčinu/matiku) — auto-detect zamieňa slovenčinu za poľštinu
- Výstup sa zapíše do JSON-u ako `media_transcript` pole

Spustenie:

```
pnpm run generate:media-transcripts anglicky-jazyk-1021        # jeden test
pnpm run generate:media-transcripts                             # všetky testy
```

github.com/openai/whisper

🎶Nahrávky

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  stdio: 'pipe',
  encoding: 'utf-8',
});
```

- **Model:** `medium` (1.5 GB) — dobrá presnosť pre vzdelávací obsah


```
    }
  ],
  "media_transcript": "Teraz si vypočujte ukážku.\nWhat kind of image did you have of Kyoto?\nI knew that Kyoto was the old capital of Japan and
},
{
  "description": "Vypočujete si druhú nahrávku. Na základe vypočutého rozhodnite o každom z tvrdení <b>08 – 13,</b> či je pravdivé <b>(A),</b> n
  "question_id": null,
  "question_type": "part",
  "media_intro": "anglicky-jazyk-B1-nahravka-2025-p2.mp3",
  "media": "anglicky-jazyk-B1-nahravka-2025-02.mp3",
```


github.com/openai/whisper

🖼️ Obrázky

```
curl -fsSL https://ollama.com/install.sh | sh

ollama pull llava:13b
```

github.com/ollama/ollama

Script volá lokálne Ollama API:

```
// projects/backend/scripts/generate-media-transcripts.ts

const response = await fetch('http://localhost:11434/api/generate', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({
    model: 'llava:13b',
    prompt: prompt,
    images: [fs.readFileSync(imagePath).toString('base64')],
    stream: false,
    options: {
      temperature: 0.1,   // nízka teplota = faktické popisy
      num_predict: 800,    // povolí dlhšie popisy
    },
  }),
});
```

- **Model:** `llava:13b` — vyváženie presnosti a rýchlosti
- **Kontext otázky sa pridá** do promptu → LLaVA vie čo je dôležité
- Výstup sa zapíše do JSON-u ako `media_description` pole
- **Cena: $0** — beží 100% lokálne

🖼️ Obrázky

```
curl -fsSL https://ollama.com/install.sh | sh

ollama pull llava:13b
```


```
  "question_id": 17,
  "image": "matematika/mat_2025_17o.png",
  "items": [
    {
      "correct_value": "24,23"
    }
  ],
  "media_description": "**1. TYPE OF IMAGE:**\n- Geometric figure (specifically a line segment)\n\n**2. VISIBLE TEXT & LABELS:**\n- A'B' is labeled as the diameter of the circle
},
```


Script volá lokálne Ollama API:

```
// projects/backend/scripts/generate-media-transcripts.ts

const response = await fetch('http://localhost:11434/api/generate', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({
    ...,
  }),
});
```

- **Model:** `llava:13b` — vyváženie presnosti a rýchlosti
- **Kontext otázky sa pridá** do promptu → LLaVA vie čo je dôležité
- Výstup sa zapíše do JSON-u ako `media_description` pole
- **Cena: $0** — beží 100% lokálne

Ollama

github.com/ollama/ollama

File Edit View Bookmarks Plugins Settings Help

```
Estimated cost: $0.000479 | Time: 5080ms
Stored explanation for question 25
Generating explanation for question 26...
Searching for embeddings: test_name="matematika-177
Found 1 chunks
First chunk preview (first 300 chars):
Test: matematika-177180
Section: Časť II

Image Description:
The image appears to be a mathematical or scientific diagra
ree-dimensional figure, possibly related to geometry or phy

RAG Quality: best=0.694, avg=0.694
Subject: math, Question Type: visual, Correct Answer:
Tokens: 1681 input + 292 output = 1973 total
Estimated cost: $0.000427 | Time: 6139ms
Stored explanation for question 26
Generating explanation for question 27...
Searching for embeddings: test_name="matematika-177180"
Found 1 chunks
First chunk preview (first 300 chars):
Test: matematika-177180
Section: Časť II

Question 27: V ovocnom sade sú stromy vysadené v rade. Medzi
il, beží od prvého stromu k druhému a naspäť, potom od prvého

RAG Quality: best=0.696, avg=0.696
Subject: math, Question Type: general, Correct Answer: D
Tokens: 1499 input + 224 output = 1723 total
Estimated cost: $0.000359 | Time: 9721ms
Stored explanation for question 27
```

Edit View Bookmarks Plugins Settings Help

```
Total related texts: 0
Found ukážka text (1083 chars)
ocessing question 39 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found ukážka text (1083 chars)
ocessing question 40 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found ukážka text (1410 chars)
ocessing question 41 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found ukážka text (1410 chars)
ocessing question 42 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found ukážka text (1410 chars)
ocessing question 43 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found ukážka text (1410 chars)
ocessing question 44 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found ukážka text (1410 chars)
ocessing question 45 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found ukážka text (1410 chars)
ocessing question 46 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found ukážka text (1410 chars)
ocessing question 47 (1 chunks)...
```

Edit View Bookmarks Plugins Settings Help

```
Total related texts: 0
Found part image description (429 chars)
ocessing question 9 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found part image description (429 chars)
ocessing question 10 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found part image description (429 chars)
ocessing question 11 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found part image description (429 chars)
ocessing question 12 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found part image description (429 chars)
ocessing question 13 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found part image description (429 chars)
ocessing question 14 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found part image description (429 chars)
ocessing question 15 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found part image description (429 chars)
ocessing question 16 (1 chunks)...
Question with Slovak ukážka context - will use shared part context
Total related texts: 0
Found ukážka text (1574 chars)
ocessing question 17 (1 chunks)...
```

3. LangChain

pnpm add langchain

pnpm add @langchain/openai

text-embedding-3-small, gpt-4o-mini

```
// projects/backend/src/components/explanations/infrastructure/clients/OpenAIClient.ts

import { ChatOpenAI, OpenAIEmbeddings } from '@langchain/openai';
import { SystemMessage, HumanMessage, AIMessage as LangChainAIMessage } from '@langchain/core/messages';

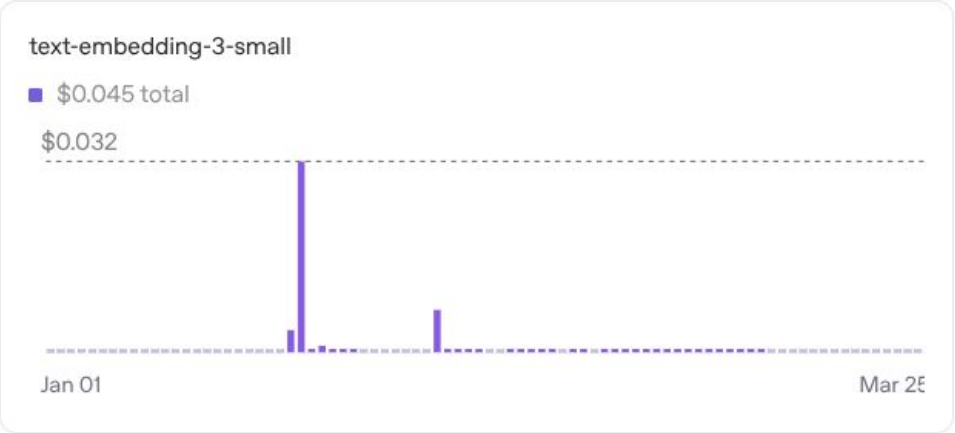
@Injectable()
export class OpenAIClient implements IAIClient {
  private readonly llm: ChatOpenAI;
  private readonly embeddings: OpenAIEmbeddings;

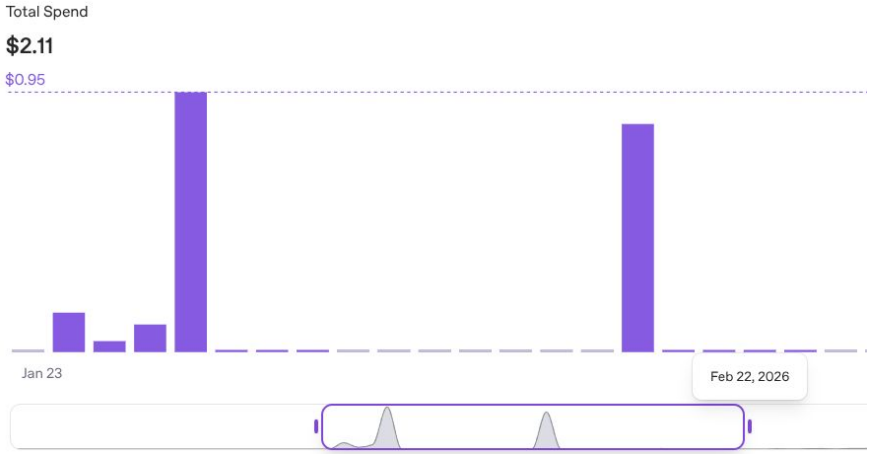
  constructor(private readonly configService: ConfigService) {
    this.llm = new ChatOpenAI({
      modelName: 'gpt-4o-mini',           // najlacnejši s dobrou kvalitou
      temperature: 0.3,                    // nízka teplota = presné odpovede
      openAIApiKey: configService.get('app.openaiApiKey'),
    });

    this.embeddings = new OpenAIEmbeddings({
      openAIApiKey: configService.get('app.openaiApiKey'),
      modelName: 'text-embedding-3-small', // 1536 dimensions
    });
  }

  async generateEmbedding(text: string): Promise<number[]> {
    return await this.embeddings.embedQuery(text);
  }

  async generateChatCompletion(messages: AIMessage[]): Promise<string> {
    const langChainMessages = messages.map((msg) => this.createMessage(msg.role, msg.content));
    const response = await this.llm.invoke(langChainMessages);
    return response.content as string;
  }
}
```


text-embedding-3-small
$0.045 total
$0.032
Jan 01
Mar 25


Total Spend
$2.11
$0.95
Jan 23
Feb 22, 2026

~ $ 2.36

```
// Initialize OpenAI Embeddings (LangChain)
const embeddings = new OpenAIEmbeddings({
  openAIApiKey: OPENAI_API_KEY,
  modelName: 'text-embedding-3-small', // 1536 dimensions, $0.02/1M tokens
});

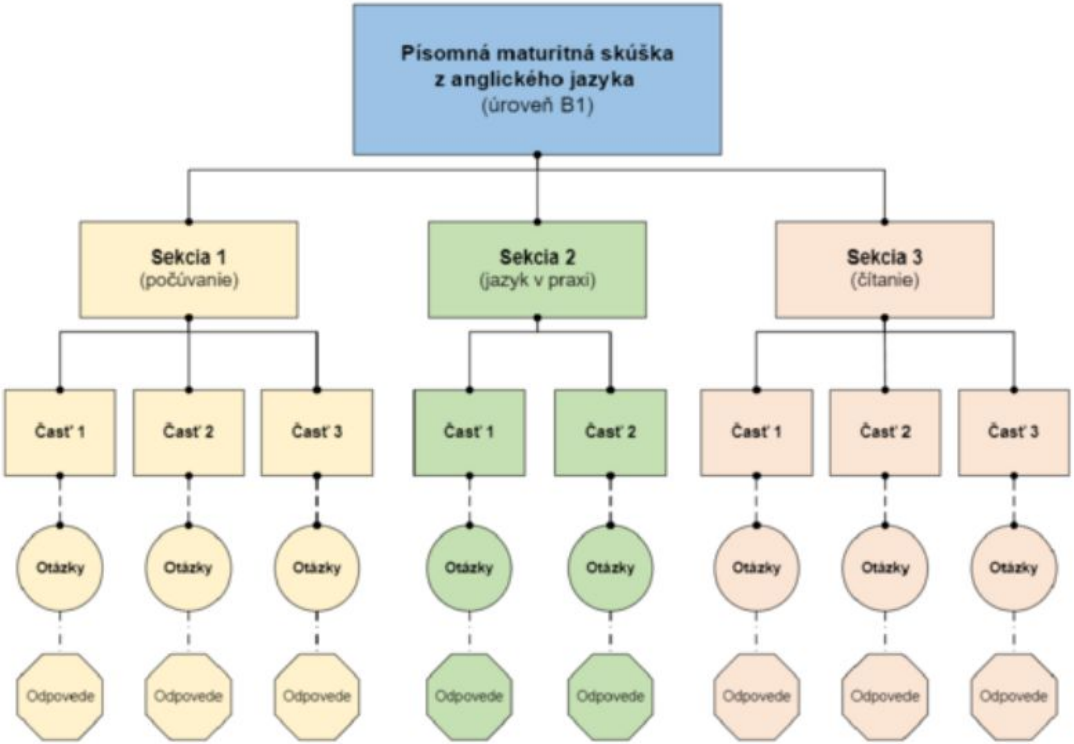
// Pre každý chunk:
const embedding = await embeddings.embedQuery(chunk);

await db
  .insert(schema.testEmbeddings)
  .values({
    testName: testName,
    question: questionId,
    chunkIndex: i,
    chunk: chunk,
    embedding: embedding,  // [0.0123, -0.0456, 0.0789, ...] — 1536 numbers
  })
  .onConflictDoUpdate({
    target: [schema.testEmbeddings.testName, schema.testEmbeddings.question, schema.testEmbeddings.chunkIndex],
    set: { chunk: chunk, embedding: embedding },
  });

// Rate limiting: 200ms medzi API volaniami
await new Promise((resolve) => setTimeout(resolve, 200));
```

Spustenie:

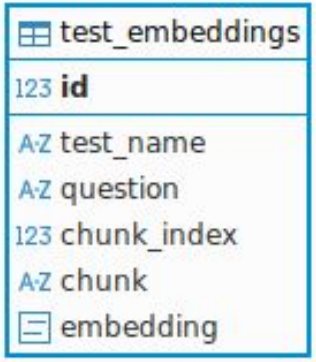
```
pnpm run generate:embeddings anglicky-jazyk-251717      # jeden test
pnpm run generate:embeddings anglicky-jazyk-251717/48    # jedna otázka
pnpm run generate:embeddings                             # všetkých 112 testov
```


Písomná maturitná skúška
z anglického jazyka
(úroveň B1)
Sekcia 1
(počúvanie)
Sekcia 2
(jazyk v praxi)
Sekcia 3
(čítanie)
Časť 1
Časť 2
Časť 3
Časť 1
Časť 2
Časť 1
Časť 2
Časť 3
Otázky
Odpovede

4. Chunking + Embedding

Typ 1 — Chunk pre otázku (chunk_index = 0)

Typ 2 — Zdieľaný kontext (reading passages, audio transcripts)



Show SQL | Enter a SQL expression to filter results (use Ctrl+Space)

Grid | Text

	123	A-Z test_name	A-Z question	123 chunk_index	A-Z chunk	embedding
1	1	anglicky-jazyk-12312	part-1-an-interviev	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[-0.0016198758,-0.0077754036,-0.011974873,-0.005014572,-0.0039478014,-0
2	2	anglicky-jazyk-12312	part-2-parenting-s	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.00087863946,0.04487331,0.019681524,-0.012051932,-0.00052497606,-0.0
3	3	anglicky-jazyk-12312	part-3-guide-to-int	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.03525088,0.040997438,-0.029690553,0.018596502,0.016095685,-0.02585
4	4	anglicky-jazyk-12312	part-2-the-abdicat	0	Test: anglicky-jazyk-123126¶Section: Section III – REA	[0.030388754,0.044791494,0.032048643,0.047115337,-0.028396886,0.03490
5	5	anglicky-jazyk-12312	part-3-the-disposa	0	Test: anglicky-jazyk-123126¶Section: Section III – REA	[0.048428033,0.036688812,-0.015507178,0.012272137,0.038280062,-0.0061
6	6	anglicky-jazyk-12312	1	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.005149793,0.0054953713,0.046592075,0.026426569,0.011620915,0.0455
7	7	anglicky-jazyk-12312	2	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.025149409,0.004413338,0.015716638,0.023979021,0.010324494,-8.25106
8	8	anglicky-jazyk-12312	3	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.030424122,0.011741029,0.016145559,0.012418143,0.0010699062,-0.0067
9	9	anglicky-jazyk-12312	4	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.009095161,0.023130268,0.033113077,0.013803547,-0.008053142,0.02341
10	10	anglicky-jazyk-12312	5	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.027257485,0.028756915,0.05039154,0.013401151,0.009947108,0.038529
11	11	anglicky-jazyk-12312	6	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.05712135,-0.012517674,0.009019886,0.031727966,0.02375466,-0.005570
12	12	anglicky-jazyk-12312	7	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.012746078,0.026884308,-0.005108938,0.011511527,-0.015970357,0.0110
13	35	anglicky-jazyk-12312	25	1	Test: anglicky-jazyk-123126¶Question 25 - Instruction:	[-0.000110965244,0.04317557,0.024977269,-0.002582464,0.013584836,0.00
14	13	anglicky-jazyk-12312	8	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.02261396,0.031067329,-0.019855868,0.040141568,-0.001468594,0.01784
15	14	anglicky-jazyk-12312	9	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.012897195,0.05298177,0.004809429,0.03014149,0.0013937437,0.020342
16	15	anglicky-jazyk-12312	10	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.012699312,0.044828095,0.0063912733,0.02166493,-0.00477116,0.031534
17	16	anglicky-jazyk-12312	11	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.0045000967,0.033149704,0.016212428,0.03184498,-0.0007980876,0.0196
18	17	anglicky-jazyk-12312	12	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.009994604,0.03476166,0.021167994,0.02550693,-0.009699907,0.0154370
19	18	anglicky-jazyk-12312	13	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.0044893213,0.06054451,-0.023023104,0.04116438,0.0105425455,0.00515
20	19	anglicky-jazyk-12312	14	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.012112201,-0.0012154734,-0.0060724597,0.01792946,0.017733153,0.001
21	20	anglicky-jazyk-12312	15	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTE	[0.01982613,0.029955283,-0.045351595,0.027740374,0.003276781,0.011783

4. Chunking + Embedding

test_embeddings

123 id

	123 chunk_index	A-Z chunk	embedding
-davenport-7-points-context	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[-0.0016198758,-0.0077754036,-0.011
nts-context	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.00087863946,0.04487331,0.019681
y-7-points-context	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.03525088,0.040997438,-0.029690
dward-viii-6-points-context	0	Test: anglicky-jazyk-123126¶Section: Section III – READING (20 points)¶	[0.030388754,0.044791494,0.032048
f-king-gillette-7-poi-context	0	Test: anglicky-jazyk-123126¶Section: Section III – READING (20 points)¶	[0.048428033,0.036688812,-0.015507
	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.005149793,0.0054953713,0.04659
		Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.025149409,0.004413338,0.0157166
		Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.030424122,0.011741029,0.0161455
		Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.009095161,0.023130268,0.0331130
	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.027257485,0.028756915,0.0503915
	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.05712135,-0.012517674,0.00901988
			2746078,0.026884308,-0.005108
	1	Test: anglicky-jazyk-123126¶Question 25 - Instructions:¶¶For questions	[-0.000110965244,0.04317557,0.0249
	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.02261396,0.031067329,-0.0198558
	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.012897195,0.05298177,0.00480942
	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.012699312,0.044828095,0.0063912
	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.0045000967,0.033149704,0.016212
	0	Test: anglicky-jazyk-123126¶Section: Section I – LISTENING (20 points)¶	[0.009994604,0.03476166,0.0211679
17 16 anglicky-jazyk-12312 11	0	Test: anglicky-jazyk-123126¶Section: Section I - LISTE	[0.0045000967,0.033149704,0.016212428,0.03184498,-0.0007980876,0.0196
18 17 anglicky-jazyk-12312 12	0	Test: anglicky-jazyk-123126¶Section: Section I - LISTE	[0.009994604,0.03476166,0.021167994,0.02550693,-0.009699907,0.0154370
19 18 anglicky-jazyk-12312 13	0	Test: anglicky-jazyk-123126¶Section: Section I - LISTE	[0.0044893213,0.06054451,-0.023023104,0.04116438,0.0105425455,0.00515
20 19 anglicky-jazyk-12312 14	0	Test: anglicky-jazyk-123126¶Section: Section I - LISTE	[0.012112201,-0.0012154734,-0.0060724597,0.01792946,0.017733153,0.001
21 20 anglicky-jazyk-12312 15	0	Test: anglicky-jazyk-123126¶Section: Section I - LISTE	[0.01982613,0.029955283,-0.045351595,0.027740374,0.003276781,0.011783

Test: anglicky-jazyk-123126
Section: Section III – READING (20 points)
Part: Part 3: The Disposable Razor of King Gillette (7 Points)

Instructions:
Read some facts from the life of King Gillette who invented the disposable razor.
Complete the senten...

5. Gen

Script `generate-stored-explanations.ts` pre každú otázku:

1. RAG retrieval (embed query → pgvector → top 5 chunkov)
2. Zostav prompt podľa predmetu + typu otázky
3. Pošli do `gpt-4o-mini`
4. Ulož vysvetlenie + metadáta (model, čas, tokeny, RAG kvalita)

```
// projects/backend/scripts/generate-stored-explanations.ts

const embeddings = new OpenAIEmbeddings({
  openAIApiKey: OPENAI_API_KEY,
  modelName: 'text-embedding-3-small',
});

const llm = new ChatOpenAI({
  modelName: 'gpt-4o-mini',
  temperature: 0.3,
  openAIApiKey: OPENAI_API_KEY,
});

// Pre každú otázku:
const ragResult = await getRelevantContext(testName, questionId, query, 5);
const systemPrompt = buildSystemPrompt(subject, questionType);
const userPrompt = buildUserPrompt(ragResult.chunks, correctAnswer, correctParagraph, questionType, subject);

const response = await llm.invoke([
  new SystemMessage(systemPrompt),
  new HumanMessage(userPrompt),
]);
```

Štruktúra promptu:

```
private buildAIPrompt(originalExplanation, contextChunks, recentMessages, newQuestion): AIMessage[] {
  return [
    { role: 'system', content: `Si pomocný učiteľ...
      ✅ Odpovedaj na VŠETKY otázky súvisiace s testom
      ❌ Odmietni len: recepty, počasie, osobné otázky, urážky
      Ak si NIE SI ISTÝ, či je otázka relevantná, ODPOVEDZ NA ŇU` },

    { role: 'user', content: `KONTEXT Z TESTU:\n\n${contextChunks.join('\n\n---\n\n')}` },

    { role: 'assistant', content: `Rozumiem, mám kontext. Pôvodné vysvetlenie: ${originalExplanation}` },

    // Posledné 2 správy študenta (konverzačný kontext)
    ...recentMessages.map(msg => ({ role: msg.role, content: msg.content })),

    // Nová otázka
    { role: 'user', content: newQuestion },
  ];
}
```

Prompting: Q

System prompt (pre angličtinu/reading):

```
function buildSystemPrompt(subject, questionType): string {
  const baseRules = `
KRITICKÉ PRAVIDLO - SPRÁVNA ODPOVEĎ:
⚠️ SPRÁVNA ODPOVEĎ JE UŽ URČENÁ V KĽÚČI ODPOVEDÍ. NIKDY ju nededukuj ani nemeň!
⚠️ Tvoja úloha je IBA VYSVETLIŤ, prečo je táto odpoveď správna.

FORMÁT ODPOVEDE:
1. Uveď správnu odpoveď (napr. "Správna odpoveď je D) ...")
2. Vysvetli PREČO v 1-2 vetách - cituj z kontextu
3. Buď STRUČNÝ a PRIAMY

KRITICKÉ PRAVIDLÁ PRE CITÁCIE:
⚠️ NIKDY nevymýšľaj citácie - použij IBA presné slová z kontextu
⚠️ Ak citáciu nemôžeš nájsť, parafrázuj BEZ úvodzoviek`;

  // Rôzne prompty pre: english/listening, english/reading, math, slovak/ukazka
  // ...
}
```

Prompting: A

User prompt — zdôraznenie správnej odpovede:

```
function buildUserPrompt(contextChunks, correctAnswer, correctParagraph): string {
  let prompt = `KONTEXT Z TESTU:\n\n${contextChunks.join('\n\n---\n\n')}`;

  prompt += `\n\n═══════════════════════════════════════════════════════════`;
  prompt += `\n⚠️ SPRÁVNA ODPOVEĎ (z kľúča odpovedi): ${correctAnswer}`;
  prompt += `\n═══════════════════════════════════════════════════════════\n`;

  if (correctParagraph) {
    prompt += `\n📍 DÔKAZ SA NACHÁDZA V ODSEKU (${correctParagraph}). Cituj z TOHTO odseku.\n`;
  }
  return prompt;
}
```

Caching

	123 id	A-Z test_name	A-Z question_id	A-Z explanation	
1	1	anglicky-jazyk-251717	1	Správna odpoveď je B) suitable size. V nahrávke sa spomína, že "if y	
2	2	anglicky-jazyk-251515	1	Správna odpoveď je C) preserving the plot of the book as much as p	
3	3	anglicky-jazyk-251717	2	Správna odpoveď je D) its political history. Kate spomína, že vedela,	
4	4	anglicky-jazyk-251717	3	Správna odpoveď je C) geographical. Kate spomína, že počas svojho	
5	5	anglicky-jazyk-251717	4	Správna odpoveď je C) give a helping hand. Kate uvádza, že "if I ha	
6	6	anglicky-jazyk-251717	6	Správna odpoveď je B) traditions and modern attitudes. V nahrávke	
7	7	matematika-251447	2	Správna odpoveď je 0,06. Hľadáme pravdepodobnosť, že súťažiaci v	
8	8	anglicky-jazyk-251649	1	Správna odpoveď je D) suggested to Warren he had a gift for writing	
9	9	anglicky-jazyk-251649	2	Správna odpoveď je B) by someone else's publisher. V nahrávke sa uvádza,	gpt-4o-mini
10	10	anglicky-jazyk-251649	41	Správna odpoveď je nightmares. V kontexte sa uvádza, že "A stubborn was	gpt-4o-mini
11	11	anglicky-jazyk-251649		že vyj	gpt-4o-mini
12	12	matematika-165178		knihy	gpt-4o-mini
13	13	matematika-165178		DEF`,	gpt-4o-mini
14	14	matematika-165178		ne, pr	gpt-4o-mini
15	15	matematika-165178	12	Správna odpoveď je 24. ¶¶Hľadáme rozdelenie čísla 110 na tri sčítance, pri	gpt-4o-mini
16	16	matematika-165178	13	Správna odpoveď je 25. ¶¶Hľadáme obvod lichobežníka, ktorý má základňu	gpt-4o-mini
17	17	matematika-165178	14	Správna odpoveď je 30. Hľadáme hodnotu `x`, pri ktorej obsah obdĺžnika, v	gpt-4o-mini
18	18	matematika-165178	15	Správna odpoveď je 49. Hľadáme minimálny počet štvorcových kúskov, kde	gpt-4o-mini

Správna odpoveď je B) by someone else's publisher. V nahrávke sa uvádza, že "When he asked for our prices, I told him that his only fee would be if he could get his publisher to publish my first novel." - preto je správna odpoveď B.

test_explanations

- 123 id
- A-Z test_name
- A-Z question_id
- A-Z explanation
- A-Z model_used
- 123 generated_by
- generated_at
- 123 generation_time_ms
- 123 tokens_used
- 123 rag_chunks_used
- 123 rag_best_similarity
- 123 rag_avg_similarity
- 123 likes_count
- 123 dislikes_count
- 123 views_count
- created_at
- updated_at

Fallback

On-demand fallback: Ak vysvetlenie ešte neexistuje, backend ho vygeneruje naživo cez rovnaký script:

```
// ScriptExplanationGenerationService — zavolá script ako child process
const { stdout } = await execAsync(
  `tsx ${this.scriptPath} "${testName}/${questionId}" --regenerate`,
  { timeout: 60000 },
);
```

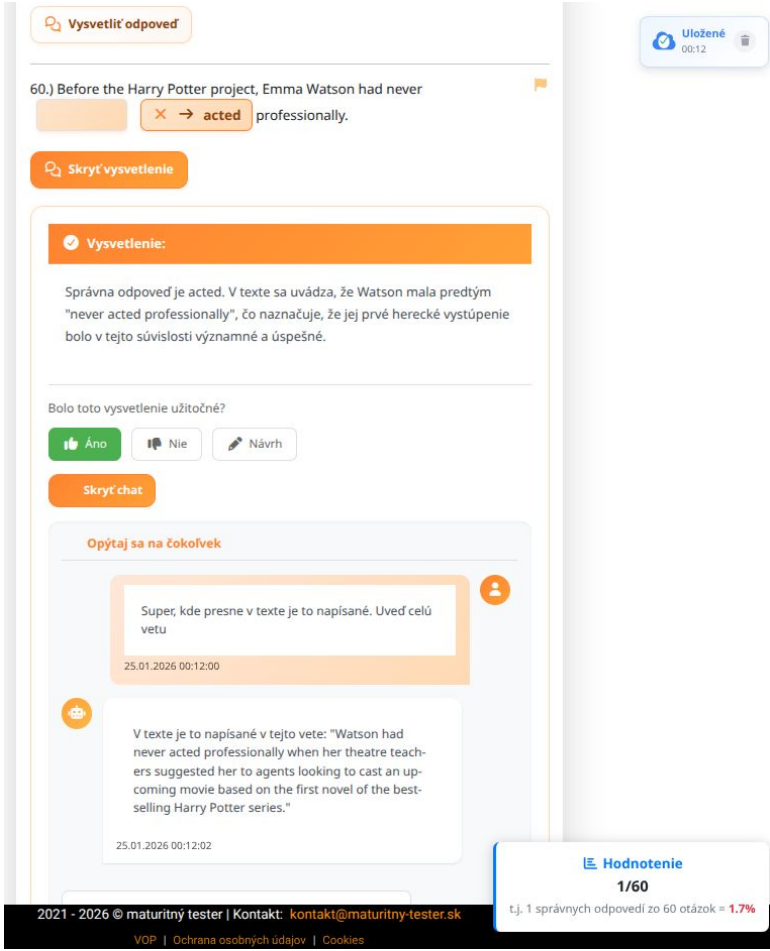
Spustenie:

```
pnpm run generate:explanations anglicky-jazyk-251717      # celý test
pnpm run generate:explanations anglicky-jazyk-251717/48    # jedna otázka
```

ZMATURUJ ĽAVOU ZADNOU!

Otestuj sa, očasuj sa, zmaturuj!

S VYSVETLENIAMI AJ AI ČETOM!



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docker Compose

2	4 GB	40 GB	€ 2.74	1/20 TB	€ 3.68/mo
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OpenAI

~ $ 2.5 (chat + embeddings)

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Ďakujem za pozornosť

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